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Your reference: EP 23945678.3
Our reference: P62047235WO-EP
Subject: European patent application titled *BALANCE CONTROL METHOD AND APPARATUS FOR WHEEL-LEGGED ROBOT, DEVICE AND STORAGE MEDIUM* in the name of Tencent Technology (Shenzhen) Company Limited

Dear Examiner,

This is in response to the communication under Rule 70(2) EPC, dated 24-02-2026. The EPO is herewith notified that the applicant wishes to proceed further with the European patent application. The Applicant herewith files two amended sets of claims (one Main Request (MR) and one Auxiliary Request (AR)) to replace the claims on file. In favour of allowability of the amended claims, the following observations are presented.

Main Request (MR)

MR-Amendments and basis
The enclosed marked-up version of the amended claims shows amendments with respect to the claims filed with entry into the regional phase before the EPO.

The independent claims have been redrafted in the two-part form.

An amended description is submitted too, in which D1-D4 has been briefly discussed in the background part. If any additional amendment to the description is deemed to be necessary by the Examiner, please directly add such amendment in the description when issuing the R71.3 EPC communication.

MR-Clarity (Art. 84 EPC)
Item 1
The Examiner objects to the clarity of feature “sliding surface” and “the state parameter value of the wheel-legged robot approaching a stable value on the sliding surface”.

Regarding the first objection, the Applicant respectfully submits that the person skilled in the art of modern robotics and control theoretical engineering (specifically non-linear sliding mode control) understands the term “sliding surface” as a well-defined technical concept. The sliding surface defines

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a surface in state space along which the controlled dynamic system exhibits desired dynamic characteristics (e.g., exponential asymptotic stability). In other words, the sliding surface represents a prescribed dynamic constraint defined in state space, designed such that when system state variables reside on or move towards this surface, the system exhibits predefined dynamic behavior (e.g., exponential asymptotic stabilization).

Furthermore, the present specification explicitly defines the structure, function, and mathematical formulation of the sliding surface. For example, paragraphs [0225]-[0256] and claims 5-7 explicitly define how the sliding surface is established, i.e. $s = \emptyset q + \Lambda\dot{\phi} + \dot{q}$.

Therefore, taken in context with the description, the term "sliding surface" is concise and clear under Article 84 EPC.

Regarding the second objection, the Examiner argues that sliding-mode control solely "keeps" parameter values stable on the sliding surface. The Applicant respectfully points out that sliding-mode control dynamics consist of two phases:

Reaching Phase (Approaching): The system states start from an arbitrary initial point and converge toward the sliding surface under the influence of the reaching law (e.g., *see* paragraph [0262] of the English application, $\dot{s} = -k_{\eta} sgn(s)$). In this phase, the state values/parameters *approach* the sliding surface/stable values.

Sliding Phase: Once on the sliding surface, the system states slide along the surface toward the equilibrium point (stable value).

Therefore, describing the state parameter value as "approaching a stable value on the sliding surface" precisely and accurately reflects both dynamic phases of sliding-mode control.

In view of the above, it is therefore submitted that the claim language is clear, concise, and technically accurate under Article 84 EPC.

MR-Novelty and Inventive step (Art. 54 EPC and Art. 56 EPC)

Item 2

The Examiner objects to the novelty of claims 1 and 12-15 over D1.

The Applicant respectfully submits that D1 fails to disclose the mapping and parameter determination features, i.e., the following features defined in claim 1:

> "*determining (1120) dynamics model parameters for the wheel-legged robot according to a dynamics motion model and the state parameter value, the dynamics model parameters defining a mapping relationship between an angular acceleration at the first time point and a rotation torque at a second time point, the angular acceleration at the first time point comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second time point comprising rotation torques of the n rotating joints*".

D1 discloses backstepping sliding-mode control for a specific 2-DOF leg-wheeled robot. In D1 (Chapter III), the controller calculates control input torques directly based on backstepping sliding-mode functions for a specific 2-DOF mechanism as follows:

D1 discloses adaptive motion control specifically designed for a 2-DOF terrain-adaptive two-wheeled mobile platform. In Section III.B of D1, D1 defines three kinematic tracking error variables: tangential error $e_1(t)$, normal error $e_2(t)$, and orientation error $e_3(t)$. Based on e_1, e_2, e_3, D1 sequentially constructs virtual control signals ϕ_1, ϕ_2, ϕ_3 (see Equation 3):

$$\phi_1 = -K_\alpha \alpha, \phi_2 = k_1 e_1 + v_r \cos e_3 , \phi_3 = \frac{k_2 \bar{e}_3 + v_r e_2 \sin e_3 + k_3 v_r \sin e_3 + \dot{\theta}_r}{1 + k_3 e_1}$$

Thereafter, D1 synthesizes two scalar control torque signals C_θ (for speed) and C_y (for heading) step-by-step using backstepping sliding mode error functions (see Equations 4 and 5):

$$C_y = g_{23}^{-1}(x)\left[-f_{23}(x) + \dot{\phi}_3 - K_{\eta o1}\text{sgn}(S_\eta) - K_{\eta o2} S_\eta\right]$$

$$C_\theta = \frac{1}{r_1 g_{22}(x) + r_2 g_{21}(x)}\left[-r_1 f_{22}(x) - r_2 f_{21}(x) - r_2 K_\alpha \dot{\alpha} + r_1 \dot{\phi}_2 - K_{S1}\text{sgn}(S_1) - K_{S2} S_1\right]$$

where $f_{21}(x), f_{22}(x), f_{23}(x), g_{21}(x), g_{22}(x), g_{23}(x)$ are specific scalar non-linear functions derived specifically for D1's 2-DOF mechanism.

Therefore, D1 computes scalar control input torques (C_θ, C_y) step-by-step through backstepping virtual control based on kinematic tracking errors e_1, e_2, e_3. However, D1 does not determine dynamic model parameters that explicitly establish a mapping matrix relationship between the angular acceleration vector at time t1 and joint rotation torques at time t2 for an arbitrary n-link generalized robot structure (n ≥ 2).

Accordingly, D1 at least fails to disclose the features "*determining (1120) dynamics model parameters for the wheel-legged robot according to a dynamics motion model and the state parameter value, the dynamics model parameters defining a mapping relationship between an angular acceleration at the first time point and a rotation torque at a second time point, the angular acceleration at the first time point comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second time point comprising rotation torques of the n rotating joints*" recited in claim 1.

It is therefore submitted that claim 1 is novel over D1.

In D1, because control signals C_θ, C_y are derived step-by-step from recursive virtual control equations tied to a specific 2-DOF tracking error model (e_1, e_2, e_3), any change in robot posture or degrees of freedom requires completely re-deriving the backstepping virtual controllers. The core motivation of using backstepping sliding-mode control in D1 is to avoid dynamic matrix inversion and parameter matrix mapping by stepping through virtual control error dynamics (e_1, e_2, e_3). A person skilled in the art attempting to adapt D1 to a generalized n-link robot would naturally follow D1's core methodology and attempt to construct higher-order recursive error equations for additional links. However, discarding D1's backstepping error formulation in favor of establishing a generalized acceleration-to-joint torque mapping parameter matrix **directly contradicts** D1's underlying design philosophy (i.e., teaches away from the present invention).

Therefore, arriving at the subject-matter of Claim 1 from D1 is non-obvious to the skilled person.

While D2 concerns manipulator end-effector disturbance observers, D3 and D4 deal with inverse dynamics or inverted pendulum mode transitions, but none discloses or teaches determining dynamic mapping parameters that establish a mapping relationship between angular accelerations and joint torques. Therefore, the skilled person starting from D1 would find no teaching or incentive in D1 or any other cited document (D2–D4) to replace D1's specific backstepping error-based controller with a dynamic model parameter mapping matrix that maps angular accelerations to joint rotation torques.

It is therefore submitted that claim 1 of the Main Request is inventive over the cited documents. The same applies *mutatis mutandis* to independent claims 12-15.

It is therefore submitted that the requirements of Article 54 and 56 EPC have been met.

Auxiliary Request (AR)

AR-Amendments and basis

The enclosed marked-up version of the amended claims shows amendments with respect to the claims filed with entry into the regional phase before the EPO.

1. Claim 1 has been amended to recite "*wherein the determining (1120) comprises: substituting the state parameter value into the dynamics motion model to determine an inertia matrix, a bias force matrix, and a gravity matrix; and processing an inverse matrix of the inertia matrix, the bias force matrix, and the gravity matrix respectively using a selection matrix to obtain an offset parameter matrix and a proportion parameter matrix, wherein the selection matrix is configured for extracting the rotation torques of the n rotating joints from the dynamics motion model; establishing (1130) n sliding surfaces according to the state parameter value and at least two sliding mode parameters, each of the n sliding surfaces corresponding to one of the n rotating joints, wherein the at least two sliding mode parameters satisfy a constraint condition of a stability criterion; calculating (1140) the rotation torques of the n rotating joints according to the n sliding surfaces and the dynamics model parameters*". Basis for the amendment can be found at least in former claims 2-3, 5 and the relevant description, such as paragraphs [0204], [0209], [0212]-[0214] the English application.

Former claim 12 has been amended in accordance with claim 1.

2. Former claims 2-3, 5 and 6 have been amended adaptively based on amended claim 1.

3. The independent claims have been redrafted in the two-part form.

An amended description is submitted too, in which D1-D4 has been briefly discussed in the background part. If any additional amendment to the description is deemed to be necessary by the Examiner, please directly add such amendment in the description when issuing the R71.3 EPC communication.

AR-Clarity (Art. 84 EPC)

Item 1

The Examiner objects to the clarity of feature "sliding surface" and "the state parameter value of the wheel-legged robot approaching a stable value on the sliding surface".

The Applicant has amended claim 1 to define that "establishing (1130) n sliding surfaces according to the state parameter value and at least two sliding mode parameters, each of the n sliding surfaces corresponding to one of the n rotating joints". The sliding surfaces are explicitly bounded by concrete joint torque mappings and sliding parameters.

Further, the term "approaching" has been eliminated and replaced by mathematically control terminology: "wherein the at least two sliding mode parameters satisfy a constraint condition of a stability criterion".

In view of the above amendments, it is submitted that the amended claims are clear and comply with the requirement of Art. 84 EPC.

AR-Novelty and Inventive step (Art. 54 EPC and Art. 56 EPC)

Item 2

The Examiner objects to the novelty of claims 1 and 12-15 over D1.

Amended claim 1 recites, among other elements, that

> **Feature I**: "*determining (1120) dynamics model parameters for the wheel-legged robot according to a dynamics motion model and the state parameter value, the dynamics model parameters defining a mapping relationship between an angular acceleration at the first time point and a rotation torque at a second time point, the angular acceleration at the first time point comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second time point comprising rotation torques of the n rotating joints*";
>
> **Feature II**: "*wherein the determining (1120) comprises: substituting the state parameter value into the dynamics motion model to determine an inertia matrix, a bias force matrix, and a gravity matrix; and processing an inverse matrix of the inertia matrix, the bias force matrix, and the gravity matrix respectively using a selection matrix to obtain an offset parameter matrix and a proportion parameter matrix, wherein the selection matrix is configured for extracting the rotation torques of the n rotating joints from the dynamics motion model; establishing (1130) n sliding surfaces according to the state parameter value and at least two sliding mode parameters, each of the n sliding surfaces corresponding to one of the n rotating joints, wherein the at least two sliding mode parameters satisfy a constraint condition of a stability criterion*".

As discussed in the Main request, neither D1 nor any of D2-D4 discloses the determination step of establishing dynamic model parameters mapping angular accelerations to joint rotation torques, i.e. Feature I of amended claim 1. The cited documents D1-D4 naturally fail to disclose any of the specific operational steps integrated into the determination step and the establishing step, i.e. Feature II of amended claim 1. In fact, D1 relies exclusively on scalar error equations (e_1, e_2, e_3) for a specific 2-DOF mechanism, which inherently avoids dynamic matrix inversion or selection matrix extraction. Therefore, D1 inevitably provides no teaching, or motivation to apply a selection matrix to decouple inverse inertia, bias force, and gravity matrices into offset and proportion matrices.

It is therefore submitted that the subject-matter of amended claim 1 of the Auxiliary request is novel.

Through the dynamics model parameters, the rotational relationship between the acceleration of the wheel-legged robot and the rotation torques of the n rotating joints can be determined. Subsequently, according to the dynamics model parameters and the sliding surfaces, the rotation torques of the n rotating joints can be quickly determined, thereby helping to reduce unnecessary conversions in the process of determining the rotation torques of the rotating joints and alleviate the computation pressure of the computer device (*see*, e.g., from paragraph [0211] of the English application).

As discussed above, none of D1-D4 discloses or hints towards the solution of the present invention, in particular the above distinguishing features, Feature I and Feature II of amended claim 1 according to the Auxiliary request. No cited document incites the skilled person to adapt D1 to arrive at such an arrangement and it is therefore not conceivable why the skilled person would have any motivation to do so. Therefore, even if starting from D1 and aiming to solve the objective technical problem of reducing real-time control torque calculation complexity, a skilled person would not find any hint to add the distinguishing features of the method followed as described in claim 1 of the Auxiliary request.

It is therefore submitted that amended claim 1 of the Auxiliary request is inventive over the cited document. The same applies *mutatis mutandis* to independent claims 12-15.

It is therefore submitted that the requirements of Article 54 and 56 EPC have been met.

Concluding remarks

By the amendments presented herewith and in view of the above argumentation, it is submitted that the claims fulfil the requirements of the EPC and are in a condition ready for allowance. Favourable

consideration of the application is therefore requested. Should there nonetheless be minor issues left outstanding, the Examining Division is kindly requested to contact the undersigned by telephone (+31 629069017) or e-mail (Shi@nlo.eu; Wang@nlo.eu).

As a precautionary measure, in the event that the Examining Division contemplates refusal of the application, Oral Proceedings under Article 116 EPC are requested.

Finally, it is noted that the present response and the enclosed submissions do not imply the abandonment of any subject matter originally filed, and the applicant reserves the right to re-instate any cancelled subject-matter either in the present application or in a divisional application.

Encl: - Amended claims – MR (marked and clean)
- Amended claims – AR (marked and clean)
- Amended description – MR and AR (marked and clean)

Yours faithfully,

/Huaizhou Shi/

Huaizhou Shi
Professional Representative